

THE **V**IRTUAL MEET EXPERIENCE

2021 - 2022

HS VIRTUAL CHALLENGE MEET #2



SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

The Virtual Challenge Meets™

2021-22 Virtual Challenge Meet #2 - Science Test

Biology Questions (B1-B20)

- B01. Enzymes such as protein kinases are responsible for the catalysis of what molecule?
A) phosphates from ATP/GTP
B) hydrogens from COOH
C) amines from proteins
D) carbon dioxide from pyruvate
- B02. The flagella and cilia are extensions of what intracellular protein?
A) actin
B) microtubules
C) collagen
D) kinesin
- B03. Organisms that have evolved to look similar despite not having a recent common ancestor are examples of
A) coevolution.
B) synapomorphies.
C) divergent evolution.
D) convergent evolution.
- B04. Mast cells are part of the innate immune response. They release numerous chemical signals including one that will increase permeability of blood vessels to white blood cells so they can move into infected tissue. What are these chemical signals?
A) antibodies
B) lysozymes
C) interferons
D) histamine
- B05. Proteins found in epithelial tissue that anchor neighboring cells together are what type of cellular junction?
A) tight
B) gap
C) desmosomes
D) plasmodesmata
- B06. If a diploid genotype is found to be: AaBBCCDd
How many different gametic genotypes would be possible?
A) 2
B) 12
C) 8
D) 16
- B07. Halophytes are:
A) plants that thrive in high saline environments
B) insects that cannot survive in high saline environments
C) bacteria that cannot survive in high saline environments
D) plants that thrive in low saline environments
- B08. Terrestrial and aquatic bacteria convert nitrogen gas into a usable form of ammonium through what metabolic process?
A) denitrification
B) deamination
C) nitrogen fixation
D) atmospheric deposition
- B09. What regulatory process is not found in prokaryotes?
A) RNA processing
B) DNA replication
C) operons
D) glycolysis
- B10. The somatic cells of an individual with Down Syndrome would be identified as
A) $2n - 1$.
B) $n + 2$.
C) $n - 2$.
D) $2n + 1$.
- B11. The inner portion of a thylakoid is called the
A) lumen.
B) inner membrane.
C) stroma.
D) matrix.

2021-22 Virtual Challenge Meet #2 - Science Test

- B12. Coenzymes NADH and FADH₂ will provide the electrons necessary for the electron transport chain to create a proton gradient. What is the proper description for the transfer of energy from the coenzymes to the ETC?
- A) NADH/FADH₂ become reduced as they pass electrons to oxidized oxygen
 - B) NADH/FADH₂ become oxidized as they pass electrons to reduced oxygen
 - C) NADH/FADH₂ are already oxidized and will become more oxidized once they pass their electrons to the ETC
 - D) NADH/FADH₂ will become oxidized as they pass their electrons to reduce water
- B13. Glycolysis and the citric acid cycle are linked by the oxidation of pyruvate into _____.
- A) pyruvate dehydrogenase
 - B) oxaloacetate
 - C) malate
 - D) Citrate
 - E) a two carbon acetate molecule
- B14. The main distinction between a protostome and deuterostome is the
- A) development of body symmetry.
 - B) creation of germ layers from diploblastic to triploblastic.
 - C) operon manipulation.
 - D) gastrulation of an embryo's mouth and anus.
- B15. The synaptic cleft is the open space found between
- A) neurons.
 - B) muscle cells.
 - C) epithelial cells.
 - D) bacterial cells.
- B16. Ribosomes are constructed of proteins and _____.
- A) RNA
 - B) polysaccharides
 - C) glycerol
 - D) lipids
 - E) monosaccharides
- B17. If the saline concentration in an environment increases, how is the water potential changed?
- A) water potential is not affected by the saline concentration
 - B) it will decrease.
 - C) it will increase.
 - D) water potential is affected by pressure change only.
- B18. In October, the WHO and CDC announced a level 3 warning of no travel to the Republic of the Congo due to an outbreak of
- A) polio.
 - B) Ebola.
 - C) MRSA.
 - D) bird flu.
- B19. Males carry one copy of the red-green colorblindness gene. This kind of inheritance pattern is known as
- A) heterozygosity.
 - B) hemizygosity.
 - C) homozygosity.
 - D) epistatic.
- B20. Which description most closely defines a community?
- A) a group of individuals that interbreed
 - B) sets of organisms and their surrounding environment
 - C) interdependent organisms that live with proximity of each other
 - D) similar organisms that cannot survive independently

- C01. What is the molar mass of nitric acid?
A) 31 g/mol D) 79 g/mol
B) 63 g/mol E) 47 g/mol
C) 48 g/mol
- C02. In the reaction
 $\text{Cd(s)} + 2\text{AgCl(s)} \rightarrow \text{Cd}^{2+} + 2\text{Ag(s)} + 2\text{Cl}^-$
A) No electrons are transferred
B) Electrons are transferred from the silver in AgCl(s) to the Cd(s)
C) The oxidizing agent is cadmium metal
D) The reducing agent is chloride ion
E) Electrons are transferred from Cd(s) to silver in AgCl(s)
- C03. How many liters of nitrogen gas can be formed if 4.50 L ammonia react with 4.50 L of oxygen from the reaction (unbalanced) below? (pressure and temperature remain constant)
 $\text{NH}_3(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
A) 4.50 L D) 2.25 L
B) 2.00 L E) 3.00 L
C) 9.00 L
- C04. What answer should be reported if 4.560 g is added to 2.6×10^{-3} g?
A) 4.563 g D) 4.5626 g
B) 4.56 g E) 7.2 g
C) 4.6 g
- C05. How many orbitals are there in $n = 3$?
A) 9 D) 5
B) 18 E) 7
C) 4
- C06. Which of the following has the shortest wavelength?
A) Green light
B) Microwaves
C) Gamma rays
D) Infrared radiation
E) Ultraviolet radiation
- C07. If 100. mL of 1.00 M barium chloride is added to 100. mL of 1.00 M potassium sulfate in a calorimeter with a known heat capacity of 1.52 kJ/K, the temperature rises by 1.73 K. What is the enthalpy change per mol of barium sulfate produced?
A) – 11.4 kJ D) – 13.2 kJ
B) – 17.6 kJ E) – 8.79 kJ
C) – 26.3 kJ
- C08. Using the reactions:
 $4\text{Bi(s)} + 3\text{O}_2(\text{g}) \rightarrow 2\text{Bi}_2\text{O}_3(\text{s})$
 $\text{Bi}_2\text{O}_3(\text{s}) + 6\text{HCl(aq)} \rightarrow 2\text{BiCl}_3(\text{aq}) + 3\text{H}_2\text{O(l)}$
If 5.00 L of 0.200 M BiCl₃ are produced, how many grams of solid bismuth were reacted?
A) 105 g D) 52.3 g
B) 209 g E) 418 g
C) 836 g
- C09. Which of the following sets of aqueous solutions will produce a precipitate when combined?
A) AgNO₃ and LiClO₄
B) Pb(NO₃)₂ and KI
C) K₂SO₄ and Cu(NO₃)₂
D) NaC₂H₃O₂ and CaCl₂
E) Ca(C₂H₃O₂)₂ and NH₄Cl
- C10. A 25.0 g sample of lead is gray in color with a density of 11.35 g/mL. It is soft, malleable, melts at 600 K and conducts electricity in the solid phase. Which property of lead is extensive?
A) Gray color
B) Density of 11.35 g/mL
C) Melting point of 600 K
D) Mass of 25.0 g
E) More than one answer is correct

- C11. Which of the following is expected to have the largest lattice enthalpy?
 A) AgCl(s) D) KCl(s)
 B) NaCl(s) E) MgCl₂(s)
 C) LiCl(s)
- C12. What is the hydroxide ion concentration in a pH 9.60 solution at 25 C?
 A) 0.64 M D) 1.5×10^{-5} M
 B) 2.5×10^{-10} M E) 1.0×10^{-14} M
 C) 4.0×10^{-5} M
- C13. The standard reduction potential for the Ag⁺/Ag electrode is +0.80 V. If the standard potential for the cell shown below is +1.24 V, what is the standard reduction potential of the Fe²⁺/Fe electrode?
 $\text{Fe} \mid \text{Fe}^{2+} \parallel \text{Ag}^+ \mid \text{Ag}$
 A) + 0.44 V D) – 2.04 V
 B) + 2.04 V E) – 0.44 V
 C) – 0.88 V
- C14. Which of the following is true?
 A) Spontaneous reactions always have $\Delta H^\circ < 0$
 B) Spontaneous reactions always have $\Delta G^\circ > 0$
 C) Spontaneous reactions always have $\Delta H^\circ > 0$
 D) Spontaneous reactions always have $\Delta S^\circ < 0$
 E) All of the statements are false
- C15. If the K_{sp} for lead(II) iodate is 2.6×10^{-13} at 25 C, what is its solubility in water at 25 C?
 A) 5.1×10^{-7} M D) 5.1×10^{-6} M
 B) 6.5×10^{-14} M E) 6.0×10^{-7} M
 C) 4.0×10^{-5} M
- C16. What is the equilibrium constant expression for the reaction (unbalanced):
 $\text{CaSO}_4(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{SO}_2(\text{g}) + \text{O}_2(\text{g})$
 A) $K_c = 1/[\text{SO}_2]^2[\text{O}_2]$
 B) $K_c = [\text{CaO}]/[\text{CaSO}_4]$
 C) $K_c = [\text{CaO}]^2[\text{SO}_2]^2[\text{O}_2]/[\text{CaSO}_4]^2$
 D) $K_c = [\text{SO}_2][\text{O}_2]$
 E) $K_c = [\text{SO}_2]^2[\text{O}_2]$
- C17. Consider the reaction
 $2 \text{A}(\text{g}) \rightarrow 4 \text{B}(\text{g}) + \text{C}(\text{g})$ rate = $k[\text{A}]$
 How long will it take for the [A] to become 1/6 of its initial value if $k = 5.20 \times 10^{-3} \text{ s}^{-1}$.
 A) 344 s D) 22.2 s
 B) 800. s E) 2070 s
 C) 57.4 s
- C18. Which of the following 1.0 M aqueous solutions would be expected to have the highest boiling point?
 A) Aluminum sulfate D) methyl alcohol
 B) Magnesium sulfate E) glucose
 C) Potassium chloride
- C19. When the mole fraction of a solute is 1.0, there is
 A) 1.0 g of solute per 100 g of solution
 B) A 1:1 ratio of solute to solvent
 C) 1.0 mole of solute and 99 moles of solvent
 D) Only solute present
 E) Only solvent present
- C20. Estimate the C – O – O bond angle in the compound, C₄H₆O₄, as shown below:

$$\text{H}_3\text{C}-\overset{\text{O}}{\parallel}-\text{O}-\text{O}-\overset{\text{O}}{\parallel}-\text{CH}_3$$

 A) < 109 D) > 120
 B) < 120 but > 109 E) = 90
 C) = 180

P01. According to Michio Kaku, the second great revolution in technology was the electric age. General Electric backed Thomas Edison and his plan to use DC electricity. Westinghouse backed _____ and his plan to use AC electricity. "In the end, Edison lost the battle and the considerable funds he invested in DC technology."

- A) Michael Faraday
- B) Guglielmo Marconi
- C) James Maxwell
- D) Nikola Tesla
- E) Joseph Henry

P02. According to Michio Kaku, "Einstein realized that the sun's gravity would be powerful enough to bend the starlight of nearby stars. Since these stars could only be seen during a solar eclipse, Einstein proposed that an expedition be sent to witness the solar eclipse of 1919 to test his theory... When this epic experiment was finally performed by astronomer _____, there was remarkable agreement between Einstein's prediction and the actual result."

- A) Georges Lemaitre
- B) Arthur Eddington
- C) Edwin Hubble
- D) Henrietta Leavitt
- E) William Herschel

P03. According to Michio Kaku, in 1930, the Solvay Conference in Brussels featured a debate for the ages. On one side were Albert Einstein and Erwin Schrodinger. On the other side were _____ and Werner Heisenberg. "Princeton physicist John Wheeler said, 'It was the greatest debate in intellectual history that I know about. In thirty years, I never heard of a debate between two greater men over a longer period of time on a deeper issue with deeper consequences for understanding this strange world of ours.'"

- A) Enrico Fermi
- B) Paul Dirac
- C) Max Planck
- D) Wolfgang Pauli
- E) Niels Bohr

P04. The German astronomer Johann Gottfried Galle discovered this planet 1846. He used the mathematical predictions made by Urbain-Jean-Joseph Le Verrier and John Couch Adams. He actually found the planet "on his first observing night, just where Le Verrier and Adams had predicted it would be."

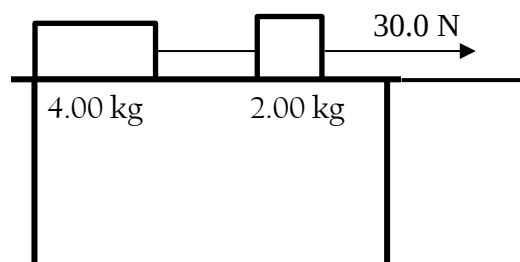
- A) Venus
- B) Mercury
- C) Neptune
- D) Uranus
- E) Pluto

P05. One terameter = _____ nanometers.

- A) 1.00×10^{15}
- B) 1.00×10^{18}
- C) 1.00×10^{21}
- D) 1.00×10^{24}
- E) 1.00×10^{27}

P06. A projectile was launched from the ground at angle of 36° above the horizontal and it hit the ground 192 m away. What was the peak height reached by the projectile?

- A) 34.9 m
- B) 37.0 m
- C) 39.1 m
- D) 41.2 m
- E) 43.3 m



P07. The coefficient of kinetic friction between the surface of the table and the wooden blocks is 0.333. Find the tension in the cord connecting the blocks shortly after a constant force of 30.0 N is applied.

- A) 17.6 N
- B) 18.8 N
- C) 20.0 N
- D) 21.2 N
- E) 22.4 N

P08. A 12.0-g bullet is fired horizontally from a 1.08-kg pistol. The muzzle speed of the bullet is 360 m/s. Find the recoil speed of the pistol.

- A) 3.34 m/s
- B) 3.56 m/s
- C) 3.78 m/s
- D) 4.00 m/s
- E) 4.22 m/s

P09. A circular saw reaches an operating speed of 1800 rpm in the process of revolving through 150 revolutions. How long does it take the saw to get up to operating speed?

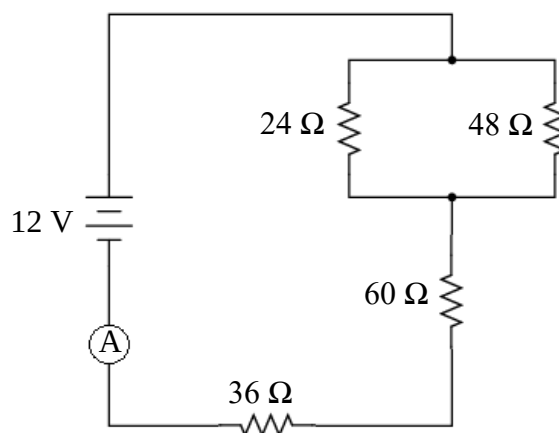
- A) 7.72 s
- B) 8.29 s
- C) 8.86 s
- D) 9.43 s
- E) 10.0 s

P10. A guitar string is 65.0 cm long and the linear density of the string is 0.001036 kg/m. What tension must the string be under if is to have a fundamental frequency of 248 Hz?

- A) 100 N
- B) 108 N
- C) 116 N
- D) 124 N
- E) 132 N

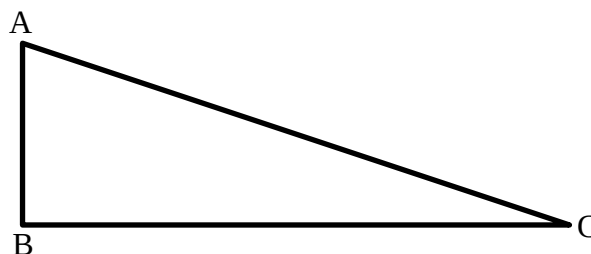
P11. A diver brought Jay a coin to see if it was made of pure gold. Jay found that the coin had a mass of 8.483 g and it displaced 0.4391 g of water. Find the density of the coin.

- A) 14,960 kg/m³
- B) 16,050 kg/m³
- C) 17,140 kg/m³
- D) 18,230 kg/m³
- E) 19,320 kg/m³



P12. Find the reading of the ammeter.

- A) 107 mA
- B) 120 mA
- C) 133 mA
- D) 146 mA
- E) 159 mA



P13. AB = 5.00 cm, BC = 12.0 cm, AC = 13.0 cm

A 100 μC charge is placed at A and at B and at C. Find the magnitude of the net electrostatic force exerted on the charge at A by the charges at B and C.

- A) 34,100 N
- B) 36,200 N
- C) 38,300 N
- D) 40,400 N
- E) 42,500 N

P14. A 20.0-m length of wire is horizontal and is located 2.00 m above the ground. If 120 A dc flows through the wire, find the value of the magnetic field produced by the current at ground level, directly under the wire.

- A) 12.0 μT
- B) 131 μT
- C) 1.42 mT
- D) 15.3 mT
- E) 164 mT

P15. A flat coil of 100 turns encloses an area of 120 cm^2 . The coil is perpendicular to a uniform magnetic field of 0.755 T that penetrates the entire area of the coil. The field is shut off and drops to zero in 188 ms . Find the average induced emf.

- A) 4.16 V
- B) 4.38 V
- C) 4.60 V
- D) 4.82 V
- E) 5.04 V

P16. A 2.00-inch-tall candle is placed 16 cm from a converging lens. If the focal length of the lens is 24 cm , what is the magnification of the image formed?

- A) 0.5
- B) 1.5
- C) 2
- D) 2.5
- E) 3

P17. Ultraviolet light with a wavelength of 240 nm is shone on a zinc target. Find the maximum speed of the emitted electrons. ($\phi = 4.31 \text{ eV}$)

- A) $110,000 \text{ m/s}$
- B) $220,000 \text{ m/s}$
- C) $330,000 \text{ m/s}$
- D) $440,000 \text{ m/s}$
- E) $550,000 \text{ m/s}$

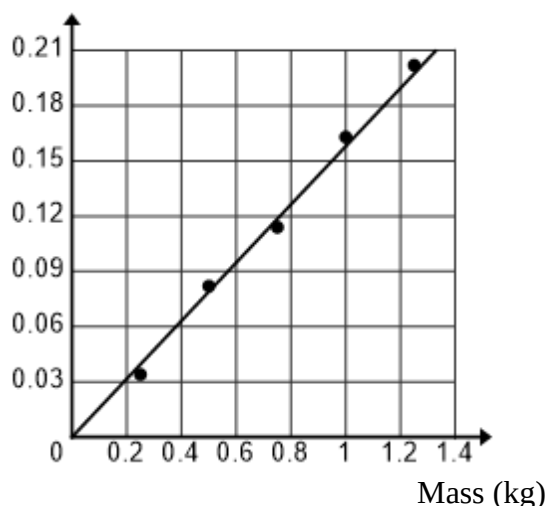
P18. ${}^{23}_{12}\text{Mg} \rightarrow {}^{23}_{11}\text{Na} + e^+ + \underline{\hspace{1cm}}$

Complete the beta decay equation.

- A) n
- B) ν
- C) p
- D) $\bar{\nu}$
- E) ${}^4_2\text{He}$



Period Squared (s^2)



P19. A mass is hung from a vertical spring and allowed to come to rest. Then it is pulled down 5.00 cm and released. The mass oscillates vertically and the period of oscillation is measured. This procedure is performed for five different masses. Use the graph above to find the spring constant of this spring. (within 0.01%)

- A) 167 N/m
- B) 210 N/m
- C) 253 N/m
- D) 296 N/m
- E) 339 N/m

P20. Monochromatic red light, $\lambda = 633 \text{ nm}$, is incident on a diffraction grating. The grating has $2000 \text{ lines per centimeter}$. A fringe pattern appears on a viewing screen 2.00 m from the grating. Find the distance from the center of the central maximum to the third bright fringe.

- A) 75.5 cm
- B) 77.7 cm
- C) 79.9 cm
- D) 82.1 cm
- E) 84.3 cm

Chemistry

Chemistry																8A 18									
1A 1		2A 2												3A 13		4A 14		5A 15		6A 16		7A 17		2 18	
1 H 1.01															5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18					
3 Li 6.94	4 Be 9.01													13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95						
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12														
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80								
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29								
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)								
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)								

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Constants

$$R = 0.08206 \text{ L}\cdot\text{atm/mol}\cdot\text{K}$$

$$R = 8.314 \text{ J/mol}\cdot\text{K}$$

$$R = 62.36 \text{ L}\cdot\text{torr/mol}\cdot\text{K}$$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

$$k = 1.38 \times 10^{-23} \text{ J/K}$$

$$h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$c = 3.00 \times 10^8 \text{ m/s}$$

$$\mathcal{R} = 2.178 \times 10^{-18} \text{ J}$$

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

Water data

$$T_{\text{mp}} = 0^{\circ}\text{C}$$

$$T_{bp} = 100^{\circ}\text{C}$$

$$C_{\text{ice}} = 2.09 \text{ J/g K}$$

$$C_{\text{water}} = 4.184 \text{ J/g K}$$

$$C_{\text{steam}} = 2.03 \text{ J/g K}$$

$$\Delta H_{\text{fus}} = 334 \text{ J/g}$$

$$\Delta H_{\text{vap}} = 2260 \text{ J/g}$$

$$K_f = 1.86\text{ }^{\circ}\text{C/m}$$

$$K_b = 0.512\text{ }^{\circ}\text{C/m}$$

2021-22 Virtual Challenge Meet #2 – Science

Free Fall Acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2 / \text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2 / \text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.00727647 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.00866492 amu
Atomic Mass Unit	amu, u	$1.66 \times 10^{-27} \text{ kg}$ 931 MeV
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2 / \text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol}\cdot\text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Electron volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Rydberg constant	R	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.103 \times 10^5 \text{ Pa}$
Horsepower	1 hp	746 W

The Virtual Challenge Meets Science Contest Answer Sheet

Conference _____

Grade Level _____

Contestant # _____

Biology

B01 _____

B02 _____

B03 _____

B04 _____

B05 _____

B06 _____

B07 _____

B08 _____

B09 _____

B10 _____

B11 _____

B12 _____

B13 _____

B14 _____

B15 _____

B16 _____

B17 _____

B18 _____

B19 _____

B20 _____

B Score

Chemistry

C01 _____

C02 _____

C03 _____

C04 _____

C05 _____

C06 _____

C07 _____

C08 _____

C09 _____

C10 _____

C11 _____

C12 _____

C13 _____

C14 _____

C15 _____

C16 _____

C17 _____

C18 _____

C19 _____

C20 _____

C Score

Physics

P01 _____

P02 _____

P03 _____

P04 _____

P05 _____

P06 _____

P07 _____

P08 _____

P09 _____

P10 _____

P11 _____

P12 _____

P13 _____

P14 _____

P15 _____

P16 _____

P17 _____

P18 _____

P19 _____

P20 _____

P Score

Grader Initials _____

OVERALL SCORE